CSE260
Assignment 02

This assignment must be hand-written. Show ALL steps in ALL questions.

Marks: 50

Q1: 5+5+5+5 = 20

Q2: 5

Q3: 5

Q4: 5

Q5: 5 + 5 + 5 = 15

1. Simplify the following boolean expressions to minimum number of literals:
   a. $(A + B)(A + \overline{B})(\overline{A} + C)$
   b. A'B'C + (A+B+C)' + A'B'C'D
   c. XY + X(Y+Z) + Y (Y+Z)
   d. (AB' (C+BD)+A'B')C

2. Find the complement of the following expression:
   (x' + y + z')(x' + y')(x + z')

3. Draw the following functions using NAND gates only:
   F(A,B,C,D) = (A'B'CD' + A'D + (B+D'))

   NB: You can't simplify the above functions and then draw using NAND gate. You have to draw based on the function given in question

4. Draw the following functions using NOR gates only:
   F(A,B,C,D) = (AB'C'D' + AD + (B+D'))

   NB: You can't simplify the above functions and then draw using NAND gate. You have to draw based on the function given in question

5. Find out SOP and POS for the following:
   a. F(A,B,C) = AB+BC'
   b. F(A,B,C,D,E) = A+ B'CD'
   c. F(V,W,X,Y,Z) = WY+WX+X'Y